Heriot-Watt University

Department of Actuarial Mathematics and Statistics

Life Insurance Mathematics B

Tutorial 3

1. (Multi-state essentials: Kolmogorov and Thiele) Consider the illness-death model in Figure 1. A life age x takes out a policy with a term of n years that pays an annuity of £1 per year continuously during any period of illness, and a sum assured of £100 on death. A premium of $\bar{P}$ per annum is payable continuously while healthy.

   Let $V^j(t)$ be the prospective policy value at age $x+t$, conditional on then being in state j $(j = 0, 1, 2)$.

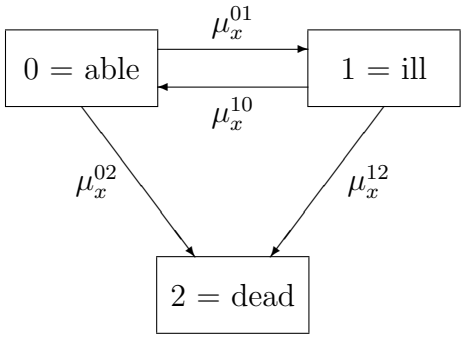


Figure 1: An illness-death model

   (a) State and prove the Kolmogorov equations for ${}_tp_x^{00}$ and ${}_tp_x^{01}$.

   (b) Write down Thiele's differential equations for this model.

   (c) State what are the boundary conditions you would use in solving Thiele's differential equations.

2. (Excel exercise: Multi-state Kolmogorov, Euler and Thiele): You can use the spreadsheet *t3q2 (Multi-state Euler Kolm and Thiele) - quick start.xlsx* to help you get started with this question. Consider the following (simplified) disability model, in which mortality has been ignored.

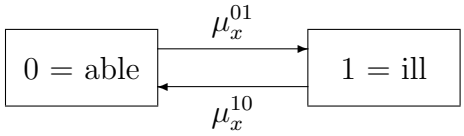


Figure 2: A simplified illness-death model

The transition intensities are given by:

$$\begin{aligned} \mu_x^{01} &= 0.01 \times 1.09^x \\ \mu_x^{10} &= 0.5. \end{aligned}$$

(a) Write down the Kolmogorov equations for ${}_tp_x^{00}$ and ${}_tp_x^{01}$.

(b) Using an Euler scheme with step size $h = 0.01$ years, solve the Kolmogorov equations to find approximate values of ${}_tp_{20}^{00}$ and ${}_tp_{20}^{01}$ for $0 \leq t \leq 10$.

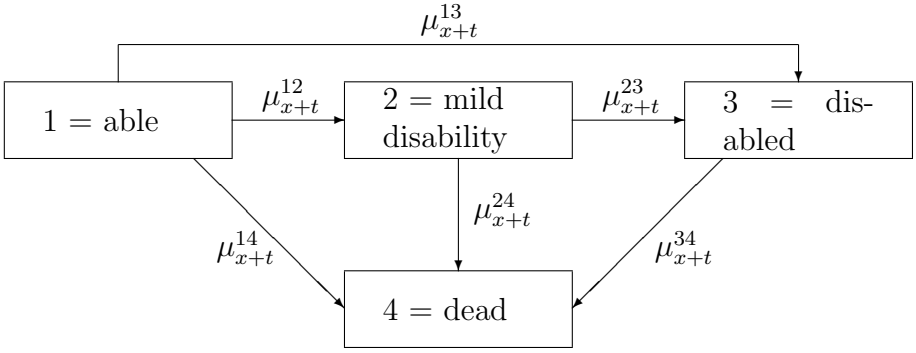
(c) An insurance company sells a 10-year disability insurance policy to someone age 20. The benefit is a disability annuity at rate £1 per annum, payable while ill. Write down Thiele's equations for the statewise policy values $V^0(t)$ and $V^1(t)$.

(d) The actuary uses the above transition intensities and a force of interest $\delta = 0.05$ per annum to calculate premiums and policy values. By trial and error, or otherwise, find the annual rate of premium such that $V^0(0) = 0$.

(e) What is the minimum value of $V^0(t)$, over all values of t? Is this good or bad?

(f) The life office decides that premiums should cease at age 29. Calculate the revised premium. Does this change your answer to (e)?

3. An insurance company uses the multiple state model shown below to calculate premiums for long-term care contracts:

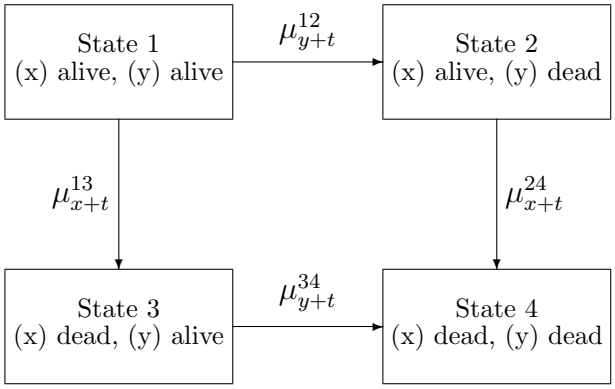


(a) Derive a differential equation for the occupancy probability ${}_tp_x^{\overline{11}}$.

(b) Show by direct substitution that that the following satisfies the differential equation derived in (a) as well as the boundary condition ${}_0p_x^{\overline{11}} = 1$:

$$ {}_tp_x^{\overline{11}} = \exp\left\{ -\int_0^t \mu^{12}_{x+r} + \mu^{13}_{x+r} + \mu^{14}_{x+r} dr \right\}. $$

4. (Short question practicing writing down Thiele DEs) A life office uses the following Markov model for pricing and valuation of joint-life insurances and annuities of all types, sold to two lives (x) and (y).



If a benefit is paid for at outset by means of a single premium, the policy value $V(t)$ is just its expected present value of the benefit at time t. Hence state Thiele's equation(s) for the following benefits.

(a) An assurance of £1 payable immediately upon the first death of (x) and (y).

(b) An assurance of £1 payable immediately upon the second death of (x) and (y).

(c) An assurance of £1 payable immediately upon the death of (x), provided (y) is then still alive.

(d) An assurance of £1 payable immediately upon the death of (x), provided (y) is then dead.

(e) An annuity payable continuously at rate £1 per annum, while (x) and (y) are both alive.

(f) An annuity payable continuously at rate £1 per annum, while at least one of (x) and (y) is still alive.

(g) An annuity payable continuously at rate £1 per annum to (y), while (y) is alive but provided (x) is dead. .

5. (Excel exercise: Thiele DE and Euler schemes for joint lives): You can start from the spreadsheet *t3q5 - quick start.xlsx*. The office in the previous question (Q 4) uses the following mortality basis:

$$\begin{aligned} \mu^{13}_{x+t} &= \mu^{24}_{x+t} = 0.0004 \times 1.09^{x+t} \\ \mu^{12}_{y+t} &= \mu^{34}_{y+t} = 0.0004 \times 1.09^{y+t} \end{aligned}$$

with force of interest $\delta = 0.05$ per annum and no expenses.

Using an Euler scheme with step size $h = 0.01$ years, find $V^1(0)$, if the office sells each of the following contracts to (40) and (35).

(a) A term assurance with sum assured £1 payable immediately on the first death of (40) and (35), if this occurs within 10 years.

(b) A term assurance with sum assured £1 payable immediately on the second death of (40) and (35), if this occurs within 10 years.

(c) A contingent assurance with sum assured £1 payable immediately on the death of (40), if this occurs within 10 years *and* (35) is still alive.

(d) An annuity payable continuously at rate £1 per annum while (40) and (35) are alive, but for a maximum term of 10 years.

(e) An annuity payable continuously at rate £1 per annum while at least one of (40) and (35) is alive, but for a maximum term of 10 years.

(f) A reversionary annuity payable continuously at rate £1 per annum while (40) is alive and (35) is dead, but for a maximum term of 10 years. (Note: *reversionary* is a term that refers to a benefit that is paid after the death of the first life. We will study reversionary annuities in more detail later in the semester.)

6. (Excel exercise: Thiele DE with trapezium rule): You can start this exercise from the Excel template *t3q6 - quick start.xlsx* which includes the necessary tables. Estimate by using Thiele's equation the single premium for a temporary contingent assurance of £50,000 payable immediately on the death of Mrs. Smith (aged 50), provided that this event occurs within 10 years and that her husband (aged 55) is alive at the date of her death. Mrs. Smith is subject to the mortality of the AF92 ultimate table and Mr. Smith is subject to the mortality of the AM92 ultimate table. An interest rate of 7.5% p.a. is to be used.

[Answer is £979.]

(Note later in the course, you will also see how to calculate the premium using an integral which is an alternative way to answer this question. The integral can be estimated in Excel by repeated application of the trapezium rule or any other suitable rule for approximate integration. The solutions on Canvas show both methods for comparison.)